AISCO HACKATHON · ALAMEDA COUNTY COMMUNITY FOOD BANK

Perishable Rescue + Equity Coordinator

Five specialized agents replace the manual decision bottleneck at a food bank — **safety-first, equity-aware, and it learns every run.**

DETERMINISTIC + LLM

DEPLOYABLE TODAY

RESCUE.INTRUDR.IO

60M

Pounds of food handled every year, across **400+ partner agencies** and **510 weekly pickups**.

95%

Of modern rescued food is **highly perishable** — fresh produce, protein and dairy, not shelf-stable cans.

TIME

The problem isn't a lack of food. **Manual sorting and routing are simply too slow** for fresh goods.

A 48-hour clock the manual desk can't beat

PHYSICAL LIMITS

Not enough warehouse cold storage to hold perishables · a severe shortage of refrigerated transport trucks.

MANUAL PROCESSING

No automated flagging for lots about to expire · one operator juggles many complex variables under time pressure.

0 h ——— 48 h

Berries and leafy greens deteriorate in **hours**. Raw chicken and meat are safe for only **1–2 days**. Without fast processing, fresh food spoils waiting on the dock.

OPERATIONAL LOSS

5–7%

of **already-rescued** food is lost — food that was paid for and delivered.

≈ 3–4 million lbs wasted annually · roughly 2–3 million meals discarded.

EQUITY FAILURE

1 in 9

182,080 Alameda County residents are food insecure (11%).

East Oakland & Fruitvale show severe need — hunger concentration traces decades-old redlining maps.

01

Habitual ordering

Agencies re-order bread and shelf-stable items out of habit.

→

02

Ignored need

Client surveys show families most want fresh produce, meat, dairy.

→

03

Mismatched delivery

The desk "honors the last order" — bread ships while fresh food expires.

→

04

Equity gap locks in

The wrong food mix quietly flows to the highest-need neighborhoods.

KEY TAKEAWAY — It's a decision bottleneck, not a food shortage. A standing order is a **flawed signal**; following it blindly locks in the wrong food mix.

STATUS QUO

One monolithic manual decision

A single operator juggles spoilage, equity, trucks *and* communication at once. Complex variables blur together — and one of them gets **silently dropped** under pressure.

OUR DESIGN

Five agents, one constraint each

Facts are handled by **deterministic code**; judgment by **LLM reasoning**. Strict boundaries mean an AI opinion **never blurs with a physical safety rule** — nothing gets silently ignored.

FACTS VS JUDGMENT · THE SPLIT THAT MATTERS

	DETERMINISTIC — MATH & PHYSICS	LLM — CONTEXT & EQUITY
ROLE	Strictly calculates reality and enforces safety.	Weighs competing human variables.
MECHANISM	Plain code — the exact same answer every time.	AI reasoning inside structured prompt guardrails.
TASKS	Spoilage math (hours-until-unsafe) · truck capacity · route feasibility.	Ranking who needs it most · correcting demand drift · multilingual notices.
RULE	**Physics wins.** Safety is never reasoned away.	**"Who needs it most"** requires real judgment.

THE FIVE-AGENT PIPELINE

01 Intake	→	02 Risk	→	03 Need	→	04 Logistics	→	05 Coordinator
clean the data		what's about to spoil?		who needs it most?		can a truck get it there?		make the plan + notify

THE LEARNING LAYER

Sharper every run

- Risk & Need agents **read past lessons before deciding**.
- The system **records new patterns** (e.g. a recurring demand drift) after every run.
- **Never crashes** — every step has a rule-based fallback. Run #1 is good; run #100 is exceptional.

Agent 1 · Intake `no AI`

INPUT

Messy raw CSVs — inventory, shelf-life, routes.

ACTION

Parses unstructured records into clean, workable data structures.

OUTPUT

Quantified lots & total pounds — `20 lots · 8,630 lbs`.

Agent 2 · Risk `deterministic math`

ACTION

Computes `(effective shelf-life - age) × 24`.

OUTPUT

Exact

HOURS-UNTIL-UNSAFE

for every lot.

OUTCOME

Auto-flags critical lots (raw meat at 0 h) with a plain-English urgency tier — an LLM explains the edge cases.

Agent 3 · Need `LLM`

Applies a
DRIFT BOOST
to correct agencies under-ordering nutritious food. Outputs a ranked 1-through-5 agency list per lot.

Agent 4 · Logistics `hard gate`

Checks cold-chain limits, depletes truck capacity sequentially, cascades to runner-up agencies — and issues a
HARD VETO
on unsafe matches.

Agent 5 · Coordinator `LLM`

Publishes the final
RESCUE RATE
and writes pickup notices in guaranteed
ENGLISH, SPANISH & CHINESE
.

PROVEN AGAINST THE STATUS QUO — COMPUTED EVERY RUN

METRIC	STATUS QUO (MANUAL DESK)	MULTI-AGENT PIPELINE
Rescue rate	71%	92% +21 pts
Spoiled on warm trucks (cold-chain ignored)	1,100 lbs	0 lbs — flagged, not lost
Delivered to highest-need agencies (need ≥ 75)	0 lbs	2,600 lbs

The status quo honors orders as written and dispatches the nearest truck — so it spoils cold food on warm routes and sends nutrition to whoever already orders it, leaving **0 lbs** for the neighborhoods furthest behind. Status quo = "honor standing orders · nearest truck · no equity or cold-chain gating".

92%

Same-day rescue rate

3,450 of 3,750 at-risk pounds placed immediately.

DRIFT CORRECTED

Steered **2,600 lbs** of fresh produce & protein to high-need East and West Oakland.

CAPACITY CASCADED

Rerouted **850 lbs** of poultry to Hayward when primary targets ran out of intake.

CONTINUOUS LEARNING

Logged spoilage patterns and recorded `A01/A03` demand drift for tomorrow's run.

A perfect 100% rescue rate would signal a **dangerous, unconstrained AI**. The 92% outcome proves the system respects absolute safety bounds.

ITEM FLAGGED	300 lbs of raw pork sausage
SAFETY RULE	Meat & poultry legally require continuous cold-chain transport
PHYSICAL REALITY	No refrigerated truck available for the assigned ZIP code
SYSTEM ACTION	**VETO** — pulled from the automated plan, flagged for manual handling

CONCLUSION — Physics wins over vibes. Safety overrules optimization.

NOT A CONCEPT — DEPLOYABLE, PRODUCTION-ORIENTED SOFTWARE

Built to plug into ACCFB's real systems, not rewritten for a demo

LIVE IN PRODUCTION

Running at rescue.intrudr.io on a real VPS — auto-deploys on every push.

NEVER FAILS ON THE MODEL

Every LLM step degrades to a rule-based fallback — **identical plan with no API**.

CHEAP & MODULAR

`~$0.001` per run · clean module boundaries so each agent can point at their live inventory / CRM.

PILOT-READY & LOW-RISK

Advisory, not autonomous. Run each morning, compare to the coordinator's plan, measure the rescue-rate delta.

RUN IT LIVE

See the five agents execute in real time

`rescue.intrudr.io`

Press **Run pipeline** — animated data flow, live agent reasoning, the status-quo contrast, and the final rescue plan.

OPEN SOURCE · JUDGED DELIVERABLE

The whole build is on GitHub

`github.com/Mossab28/perishable-rescue`

Five agents · LangGraph orchestrator · learning layer · live SSE demo · real USDA + Alameda data. README, architecture & data sources included.

92% RESCUE RATE | +21 PTS VS STATUS QUO | DEPLOYABLE